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|-----------------------------------------------------|-----------------------------------------------|
| Program : <b>Diploma in Automation and Robotics</b> |                                               |
| Course Code : <b>6332A</b>                          | Course Title: <b>Renewable Energy Systems</b> |
| Semester: <b>6</b>                                  | Credits: <b>4</b>                             |
| Course Category : <b>Open Elective</b>              |                                               |
| Periods per week: <b>4 (L:4 T:0 P:0)</b>            | Periods per semester: <b>60</b>               |

### Course Objectives:

- To understand the renewable energy sources.
- To understand the working of power plants using renewable energy.
- To understand the solar power plants.
- To understand the various methods of energy conversion from solar.
- To understand the wind power plants.
- To comprehend the components in wind electrical systems.
- Applications of AI and Robotics in Energy Sector

### Course Prerequisites:

| Topic                                      | Course code | Course name                                                                         | Semester |
|--------------------------------------------|-------------|-------------------------------------------------------------------------------------|----------|
| Basics of electrical networks and machines |             | Fundamentals of Electrical circuits and machines , Electrical circuits and Machines | 2,3      |

### Course Outcomes:

On completion of the course, the student will be able to:

| CO <sub>n</sub> | Description                                                                                  | Duration (Hours) | Cognitive Level |
|-----------------|----------------------------------------------------------------------------------------------|------------------|-----------------|
| CO1             | Understand the renewable energy sources and \working of power plants using renewable energy. | 16               | Understanding   |
| CO2             | Understand the solar power plants and various methods of energy conversion from solar..      | 14               | Understanding   |
| CO3             | Understand the wind power plants and components in wind electrical systems                   | 16               | Understanding   |

|     |                                                                     |    |                            |
|-----|---------------------------------------------------------------------|----|----------------------------|
| CO4 | To understand the applications of AI and Robotics in Energy Sector. | 12 | Understanding and Applying |
|     | Series Test                                                         | 2  |                            |

### CO – PO Mapping

| CO  | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 |
|-----|-----|-----|-----|-----|-----|-----|-----|
| CO1 | 3   |     |     |     |     |     |     |
| CO2 | 3   | 2   |     |     |     |     |     |
| CO3 |     |     | 3   |     |     |     |     |
| CO4 | 3   |     |     |     |     |     |     |

3-Strongly mapped

2-Moderately mapped

1-Weakly mapped

### Course Outline

| Module Outcomes                                                                                                                                                                                                                                                                                                                                                   | Description                                                                                                                   | Duration (Hours) | Cognitive Level |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|------------------|-----------------|
| CO1                                                                                                                                                                                                                                                                                                                                                               | <b>Understand the renewable energy sources and working of power plants using renewable energy</b>                             |                  |                 |
| M1.01                                                                                                                                                                                                                                                                                                                                                             | Explain different forms of non-conventional energy sources. Basics of solar, bio-gas, wind, tidal, geothermal energy sources. | 4                | Understanding   |
| M1.02                                                                                                                                                                                                                                                                                                                                                             | Explain the geothermal and hydrothermal resources.                                                                            | 2                | Understanding   |
| M1.03                                                                                                                                                                                                                                                                                                                                                             | Explain the methods of ocean, thermal power generation                                                                        | 3                | Understanding   |
| M1.04                                                                                                                                                                                                                                                                                                                                                             | Describe different types of fuel cells- hydrogen, fossil fuel, hydrocarbon fuel, alcohol fuel,hydrazine fuel.                 | 4                | Understanding   |
| M1.05                                                                                                                                                                                                                                                                                                                                                             | Describe the open cycle and closed MHD generation system.                                                                     | 3                | Understanding   |
| <b>Contents:</b> Non conventional energy sources – solar – bio gas – wind – tidal – geo thermal – bio gas – conversion – factors effecting bio digestion – classification – dom and drum type – methods of energy extraction from biomass – classification of bio mass gasifiers. Resources - geo thermal – hydrothermal – vapour dominated and liquid dominated- |                                                                                                                               |                  |                 |

problems related to geo thermal – interconnections geo thermal fossil systems – comparison – geo thermal preheat hybrid system- fossil superheat hybrid system – applications geo thermal – power generation – ocean thermal – open cycle OTEC- closed cycle OTEC. Tidel power – basic principle – schematic layout – single basin tidal system - Chemical energy – operation and classifications of fuel cells - hydrogen- fossil fuel- hydro carbon fuel- alcohol fuel – MHD power plants.

|                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                   |   |               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|---|---------------|
| <b>CO2</b>                                                                                                                                                                                                                                                                                                                                       | <b>To understand the solar power plants and various methods of energy conversion from solar.</b>                  |   |               |
| M2.01                                                                                                                                                                                                                                                                                                                                            | To explain the solar radiation received on the earth's surface , solar radiation geometry and measurement         | 3 | Understanding |
| M2.02                                                                                                                                                                                                                                                                                                                                            | To describe the principles of the conversion of solar radiation into heat and different types of solar collectors | 3 | Understanding |
| M2.03                                                                                                                                                                                                                                                                                                                                            | Understand solar energy storage systems, solar pond                                                               | 3 | Understanding |
| M2.04                                                                                                                                                                                                                                                                                                                                            | Understand applications of solar energy in heating                                                                | 2 | Understanding |
| M2.05                                                                                                                                                                                                                                                                                                                                            | Understand thermal electric conversion and photoelectric conversion                                               | 3 | Understanding |
|                                                                                                                                                                                                                                                                                                                                                  | Series Test – I                                                                                                   | 1 |               |
| <b>Contents:</b> Solar energy – solar radiation – measurements - solar energy conversion in to heat – solar collectors - focusing – storage systems – solar pond - Application of solar energy in heating - Thermal electrical conversion of solar energy. Photo electrical energy conversion – principle – construction of PV modules– storage. |                                                                                                                   |   |               |
| <b>CO3</b>                                                                                                                                                                                                                                                                                                                                       | <b>Understand the wind power plants and components in wind electrical systems</b>                                 |   |               |
| M3.01                                                                                                                                                                                                                                                                                                                                            | To describe the principles of wind energy conversions and components of WEC                                       | 4 | Understanding |
| M3.02                                                                                                                                                                                                                                                                                                                                            | To explain wind energy collectors and different types of wind machines                                            | 4 | Understanding |
| M3.03                                                                                                                                                                                                                                                                                                                                            | To describe the components of wind electric systems and different schemes of electric generation                  | 4 | Understanding |
| M3.04                                                                                                                                                                                                                                                                                                                                            | To describe wind energy storage and various applications of wind energy.                                          | 4 | Understanding |
| <b>Contents:</b> Wind energy conversion – components in WEC system- classification-comparison - merits and demerits- Wind electric generation – classifications – storage – application - Environmental aspects of wind energy.                                                                                                                  |                                                                                                                   |   |               |

|                                                                                                                                                                                                                                                                                                                                       |                                                                                |          |               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|----------|---------------|
| <b>CO4</b>                                                                                                                                                                                                                                                                                                                            | <b>To understand the applications of AI and Robotics in Energy Sector.</b>     |          |               |
| M4.01                                                                                                                                                                                                                                                                                                                                 | Use AI and Machine Learning for prediction for Power theft and fraud detection | 3        | Understanding |
| M4.02                                                                                                                                                                                                                                                                                                                                 | Use of data analytics for customer engagement                                  | 3        | Understanding |
| M4.03                                                                                                                                                                                                                                                                                                                                 | Use of block chain for energy trading and energy storage                       | 2        | Applying      |
| M4.04                                                                                                                                                                                                                                                                                                                                 | Use of prediction for grid management, efficiency and security                 | 4        | Applying      |
|                                                                                                                                                                                                                                                                                                                                       | <b>Series Test – II</b>                                                        | <b>1</b> |               |
| <b>Contents:</b> Use of AI and machine learning for prediction Use of ML and AI for reduce Power Theft and Energy Fraud Detection. Use of data analytics for customer engagement-energy usage reduction, , Usage of block chain for energy trading and energy storage, Use of prediction for grid management, efficiency and security |                                                                                |          |               |

#### Text / Reference

| <b>T/R</b> | <b>Book Title/Author</b>                                                                    |
|------------|---------------------------------------------------------------------------------------------|
| T1         | D.S. Chauhan. Non-Conventional Energy Resources: New age International                      |
| T2         | G.D. Rai. Non-conventional Energy Sources: Dhanpath Rai & Sons                              |
| R1         | Text book of renewable energy S C Bhatiya and R K Gupta , WPI                               |
| R2         | Fundamentals Of Renewable Energy Systems D. Mukherjee ,S Chakrabarthi New Age International |

#### Online Resources

| <b>Sl.No</b> | <b>Website Link</b>                                                                                                                                                                            |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1            | <a href="https://onlinecourses.nptel.ac.in/noc21_ge04/preview">https:// onlinecourses.nptel.ac.in/noc21_ge04/preview</a>                                                                       |
| 2            | <a href="https://energydigital.com/top10/top-10-applications-of-AI-and-Robotics-in-Energy-Sector">https:// energydigital.com/top10/top-10-applications-of-AI-and-Robotics-in-Energy-Sector</a> |